Graphical Models, COMPGI08/COMPM056, 2017-2018

Answer all THREE questions.

Marks for each part of each question are indicated in square brackets

Calculators are NOT permitted

1. a. i. For a distribution $p(x_1,\ldots,x_N)$ on discrete random varibles, explain what is meant by two variables x_i and x_j being dependent.

[2 marks]

Bookwork.

ii. Describe what is meant by a belief network and why its graphical representation must be a Directed Acyclic Graph (DAG).

[2 marks]

Bookwork

iii. Explain why any distribution can be written as a belief network.

[2 marks]

$p(x_1,x_2,\ldots,x_n) = p(x_1)p(x_2|x_1),\ldots,p(x_n|x_1,\ldots,x_{n-1})$

iv. Explain why the lack of an edge in a DAG implies some independence statement in all distributions consistent with the DAG factorisation. Explain also why the presence of an edge in a DAG does not imply a dependence statement in all distributions consistent with the DAG factorisation.

[3 marks]

Any example of an implied indepence statement is sufficient for the first part. Any example of a distribution with independence statement not implied by the graph struture is sufficient for the second part.

b. i. Explain what is meant by a Markov network.

[2 marks]

Bookwork.

ii. Show that any distribution can be written as a Markov network.

[3 marks]

Clique graph on $(x_1,\ldots,x_N)$ generates such representation.

iii. Consider a belief network whose DAG has adjacency matrix A. Now form an undirected graph with adjacency matrix B containing elements $B_{ij} = 1$ if either $A_{ij} = 1$ or $A_{ji} = 1$; $B_{ij} = 0$ otherwise. Can every belief network with DAG A be represented by a Markov network with adjacency matrix B? If so, explain why. If not, give a counter-example.

[3 marks]

$A -> B <- C$ is a counter-example.

c. i. Explain what is meant by saying that two graphs are "Markov equivalent"

[2 marks]

The set of independence statements expressed by the two graphs are the same.

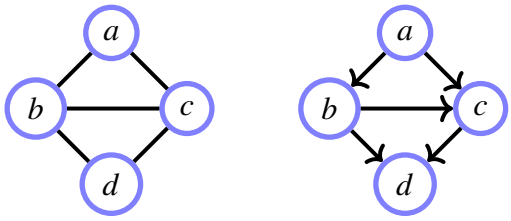
ii. Describe an algorithm to determine if two graphs are Markov equivalent.

[2 marks]

Define the skeletons and immoralities of the two graphs. If they are both the same, the graphs are Markov equivalent.

iii. Explain if the following two graphs are Markov equivalent.

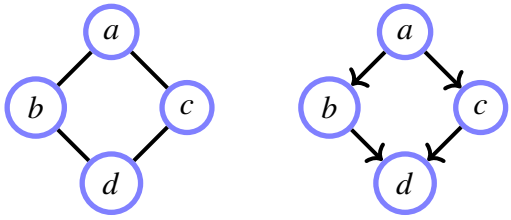
[3 marks]



They are Markov equivalent. There are no immoralities and their skeletons are the same.

iv. Explain if the following two graphs are Markov equivalent.

[3 marks]



They are not Markov equivalent. Their immoralities are not the same.

[Total 27 marks]

2. a. Describe a sum-product algorithm for computing marginals in a factor graph. Explain which factor graphs the algorithm is applicable to.

[4 marks]

Bookwork, the answer is in the lecture notes.

b. The Hidden Markov Model defines a distribution on observations $v_{1:T}$ and 'hidden' variables $h_{1:T}$:

$$p(v_{1:T}, h_{1:T}) = p(v_1|h_1)p(h_1)\prod_{t=2}^{T} p(v_t|h_t)p(h_t|h_{t-1})$$

i. Derive a recursive algorithm for computing $p(h_t|v_{1:t})$.

[3 marks]

This is bookwork. Should take about 5 minutes to answer.

ii. Derive a recursive algorithm for computing $p(h_t|v_{1:T})$.

[3 marks]

This is bookwork. Should take about 5 minutes to answer.

iii. Explain how to efficiently compute $p(v_{1:T})$.

[5 marks]

This is bookwork. Should take about 5 minutes to answer.

iv. Explain how to compute the prediction of an (as yet) unobserved future observation $p(v_{T+1}|v_{1:T})$.

[5 marks]

This is bookwork. Should take about 5 minutes to answer.

v. Explain how to compute $p(h_t, h_{t+1}|v_{1:T})$.

[2 marks]

This is bookwork. Should take about 5 minutes to answer.

vi. Derive an algorithm to efficiently compute $\arg\max_{h_{1:T}} p(h_{1:T}|v_{1:T})$.

[3 marks]

Bookwork.

c. i. Explain what is an EM algorithm. Explain what the Em and the M steps of the algorithm are and show why the likelihood is non-decreasing when running EM algorithm.

[3 marks]

Bookwork. The answer is in the lecture slides.

ii. Given a set of training observations $\mathcal{V} = \left\{v_{1:T}^n, n = 1,\ldots,N\right\}$ we wish to learn the parameters of the transition matrix A and emission matrix B of the Hidden Markov Model

$$p(h_t = i|h_{t-1} = j) = A_{ij}, \qquad i = 1,\ldots,H, \quad j = 1,\ldots,H$$

$$p(v_t = k|h_t = i) = B_{ki}, \qquad k = 1,\ldots,V, \quad i = 1,\ldots,H$$

Derive both the E and the M steps of the Expectation Maximization algorithm for maximising the likelihood $p(\mathcal{V}|A,B)$ with respect to A and B.

[6 marks]

Bookwork.

iii. Explain whether it is possible to sample efficiently (that is form a 'perfect' sample) from a joint set of variables $h_{1:T}$ from the distribution $p(h_{1:T}|v_{1:T})$. If you think this is possible, derive such an algorithm, otherwise explain why no efficient algorithm exists.

[7 marks]

The student should recognise that conditioning on v gives an undirected chain in h. This can then be transformed into a directed chain by using for example the sum-product algorithm to find $p(h_t, h_{t+1}|v_{1:T})$. This can then be re-expressed as $p(h_{t+1}|h_t, v_{1:T})$, for which one can then simply apply forward sampling to make an exact sampler.

[Total 38 marks]

3. a. i. Explain what a KL-divergence between two distributions is.

[5 marks]

Bookwork

ii. Suppose that one wants to approximate a bimodal distribution P (for example, a mixture of two gaussians) with a unimodal distribution Q (for example, a gaussian). Explain what would be the difference between the best approximations if one uses $KL(P||Q)$ and $KL(Q||P)$.

[5 marks]

The former one will result in a distribution concentrated in one of the modes of P whereas the latter approximation will try to cover both.

b. i. Explain the procedure of rejection sampling. What is the acceptance rate? What are its pros and cons?

[5 marks]

Suppose that we know that $p(x) < cq(x)$ for all x and we can sample from $q(x)$ efficiently. Then we can sample from $q(x)$ and accept the sample with probability $\frac{p(x)}{cq(x)}$. This gives independent samples, but finding a good distribution $q(x)$ with small c and hence high acceptance rate is hard, especially in high dimensions.

ii. Explain the procedure of MCMC sampling and what is meant by a detailed balance condition.

[5 marks]

Bookwork, the answer is in the lecture notes.

iii. Explain what Gibbs sampling is. Show that Gibbs sampling satisfies detailed balance condition.

[5 marks]

Bookwork, the answer is in the lecture slides.

iv. Explaing the procedure of Metropolis-Hastings sampling. Show that it satisfies detailed balance conditions.

Bookwork, the answer is in the lecture slides.

[5 marks]

[Total 32 marks]

END OF PAPER